Bruce Markman

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Solutions engineer who leads wildfire risk modeling projects from scoping through delivery, engineers custom tools for sales and model development, and guides clients from fire modeling results to mitigation decisions.

EDUCATION

Master of Science (M.S.) in Geography — 2023 - 2025

San Diego State University

Thesis: Estimating Residual Dry Matter across Conservation Grazing Lands with Field Spectroscopy and UAV LiDAR
Concentration: GIScience and Remote Sensing
Activities: American Society for Photogrammetry and Remote Sensing President (2024-25)

Bachelor of Science (B.S.) in Conservation Biology — 2018 - 2022

University of California - Davis

Minor: Geographic Information Systems
Activities: The Wildlife Society Chapter Representative (2019-2022)

PROFESSIONAL EXPERIENCE

Solutions Engineer — May 2026 – Present

Xyloplan — **San Diego, CA**

- Lead client projects end to end, from scoping and kickoff through delivery, managing timelines and presenting fire modeling results and mitigation recommendations to stakeholders
- Partner with sales as a sales engineer to optimize the pipeline, building custom demos, risk assessments, and tools that show prospects their wildfire risk
- Configure, run, and QA/QC Fire Pathways and XFM fire behavior simulations to generate burn probability, fire intensity, and ember exposure outputs used in mitigation planning
- Build custom Python geospatial scripts (GeoPandas, Rasterio, Shapely, GDAL/OGR) to classify structures by wildfire risk from FBFM40 and ember data, delineate work units, and automate QA/QC across the fire modeling pipeline
- Collaborate with software engineers and fire scientists on model development, data standards, and internal tools, contributing pull requests and participating in code review

GIS Analyst — September 2025 – March 2026

Chen Ryan Associates — **San Diego, CA**

- Conducted GIS management, QA/QC, and mapping/web applications for General Plan Updates, Safe Routes to School, and transportation planning projects across Southern California

Graduate Research Assistant — August 2023 – June 2025

San Diego State University — **San Diego, CA**

- Collaborated with The Nature Conservancy (TNC) ecologists and NASA researchers on field-calibrating and validating hyperspectral imagery and UAV LiDAR to monitor residual dry matter across California Rangelands for grazing and wildfire fuels management at Dawson Los Monos Canyon Reserve (UCSD) and Jack & Laura Dangermond Preserve (TNC). This

project led to a peer-reviewed publication in the journal *Remote Sensing: https://doi.org/10.3390/rs17142352*

- Served as President of the American Society of Photogrammetry and Remote Sensing (ASPRS) and led the acquisition of over $50,000 in grant funds to support student-professional networking activities in the geospatial and environmental research fields

Natural Resources Management Intern (GIS) **May – August 2024**
City of San Diego **San Diego, CA**

- Supported the San Diego Management and Monitoring Program with Inspect and Manage rare plant field surveys and ESRI ArcGIS Enterprise database management across Open Space properties

Research Fellow **June – September 2022**
National Center for Sustainable Transportation (NCST) **Davis, CA**

- Collaborated with scientists at UC Davis and Universidad Complutense de Madrid to study impacts of traffic noise and light on Mountain Lion (*Puma concolor*) movement, managing camera traps, wildlife databases, and geospatial visualization at remote field sites

Seasonal GIS Analyst **December 2020 – August 2021**
San Dieguito River Park JPA **San Diego, CA**

- Mapped *Arundo donax* in the park with UAV imagery to determine invasive species spread after 2010-2012 treatment by the California Invasive Plant Council
- Initiated and completed burn scar analysis from the 2007 Witch Creek fire using satellite imagery to understand wildfire spatial patterns and impacts on native vegetation

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AWARDS & HONORS

SDSU Master's Research Scholarship, McFarland Geography Scholarship, Richard Wright Award in Cartography, Pitt Virginia Warner Endowed Scholarship, and William and Vivian Finch Scholarship in Remote Sensing (2024); Oren Pollak Memorial Research Fund, The Nature Conservancy (2023); National Center for Sustainable Transportation Research Fellow (2022)

TECHNICAL SKILLS

Fire Behavior & Wildfire Modeling: Fire Pathways, XFM, LANDFIRE datasets (EVT, FBFM40, CBH, FDist), burn probability and fire intensity analysis, structure and ember risk classification

Programming & Geospatial Scripting: Python (GeoPandas, Rasterio, Shapely, GDAL/OGR, ArcPy), SQL, Git, JavaScript

Certifications: FAA Part 107 Certified Drone Pilot, S-190 Introduction to Wildland Fire Behavior, Using Spatial Data in AI Workflows (LinkedIn), AI Fluency (Anthropic), IT Automation with Python (Google)

GIS & Spatial Software: QGIS, ESRI Suite (ArcGIS Pro, Field Maps, Experience Builder), ENVI, Google Earth Engine, LP360, LAStools, TransCAD

Remote Sensing Technologies: UAV Operating and Imaging, Photogrammetry, UAV LiDAR, Imaging Spectroscopy, Satellite imagery acquisition and processing

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LANGUAGES

English, Russian (heritage), **Spanish** (elementary)